MATH 111 Calculus 1

Limits at infinity

Horizontal asymptotes

Week 2 · Lecture 6 · Stewart §3.4

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. In your head: what is $\dfrac{x^2-1}{x^2+1}$ for $x = 10$? For $x = 100$? Where is it heading?

Two questions from Lecture 5

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"Is the composition of two discontinuous functions always discontinuous?"

No. Let $g(x) = 1$ for $x \geq 0$ and $g(x) = 0$ for $x < 0$ (a jump at 0), and take $f = g$. Then $g(x)$ is always 0 or 1, and $f(0) = f(1) = 1$, so $f(g(x)) = 1$ for every x — a constant, as continuous as they come. The theorems go one way only.

STAND UP

"A graph of a function can never cross its horizontal asymptote."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

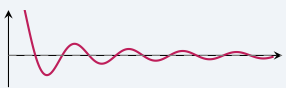
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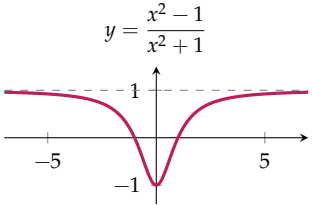
Verdict

FALSE. $y = \dfrac{\sin x}{x}$ has the horizontal asymptote $y = 0$ and crosses it at every multiple of π — infinitely often. An asymptote describes where the graph is heading as $x \to \infty$; it is not a fence. (Most of us learned the fence version at school. Today we replace it by a definition.)



What happens far away?

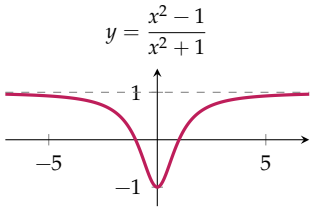
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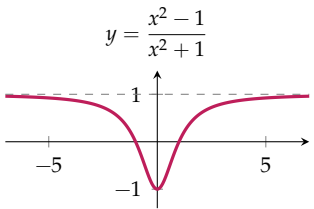


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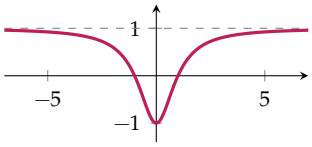
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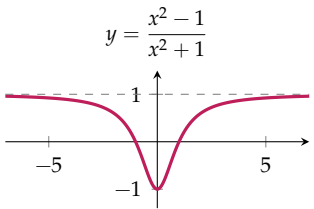
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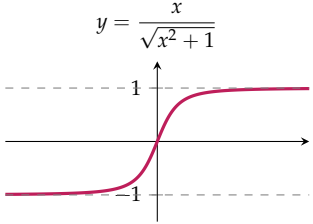
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"$x \to \infty$" reads "x grows without bound" — ∞ is not a number, and x never arrives.

Horizontal asymptotes

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The line $y = L$ is a horizontal asymptote of $y = f(x)$ if $\lim_{x \to \infty} f(x) = L$ or $\lim_{x \to -\infty} f(x) = L$.



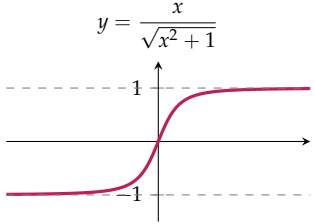
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$g(x) = \dfrac{x}{\sqrt{x^2+1}} \to 1$ as $x \to \infty$ and $\to -1$ as $x \to -\infty$.



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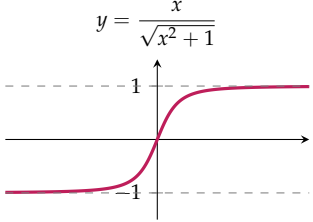
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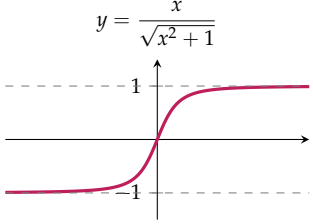
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All of today rests on this fact and the limit laws (valid for $x \to \pm\infty$ as for $x \to a$).



Rational functions: divide by the highest power

Example. $\lim_{x\to\infty} \dfrac{3x^2 - x - 2}{5x^2 + 4x + 1}$ — substitution gives "∞/∞", which is not a number.
Rewrite first.

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For $x \to -\infty$ the same steps give $\frac{3}{5}$ again: $y = \frac{3}{5}$ is the horizontal asymptote in both directions.

Question 12

$$\lim_{x\to\infty}\left(\sqrt{x^2+x}-x\right) = \ ?$$

A. 0 — for large x, $\sqrt{x^2+x}$ is practically x.

B. $\dfrac{1}{2}$

C. ∞ — both terms grow without bound.

D. It does not exist: $\infty - \infty$ is undefined.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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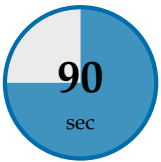
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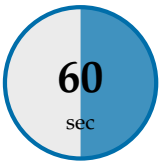
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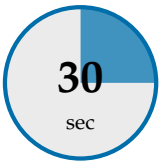
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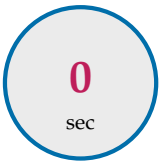
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Hands down. Thank you.

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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Answer

B. Multiply by the conjugate:

$$\sqrt{x^2+x}-x = \frac{x^2+x-x^2}{\sqrt{x^2+x}+x} = \frac{x}{\sqrt{x^2+x}+x} = \frac{1}{\sqrt{1+1/x}+1} \to \frac{1}{1+1} = \frac{1}{2}.$$ "Practically x"

(A) is true, but the difference is what we are asked about, and it is not practically 0.

$\infty - \infty$ is not undefined and not 0: it is an instruction to rewrite — like $0/0$.

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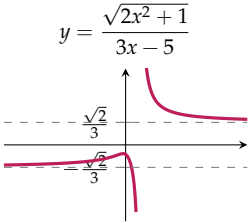
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Remember

Like $0/0$ in Lecture 2, the forms $\infty - \infty$ and ∞/∞ tell you nothing yet — they tell you to rewrite, then take the limit. Square roots: the conjugate. Quotients: divide by the highest power.

Square roots and $x \to -\infty$: the sign trap

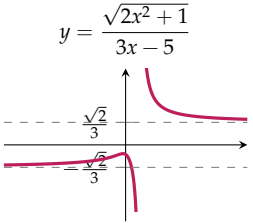
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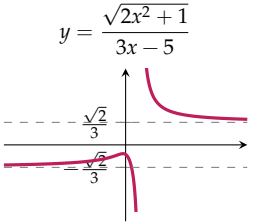


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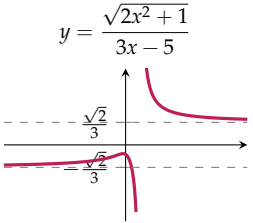


Square roots and $x \to -\infty$: the sign trap

Example. $\displaystyle\lim_{x\to-\infty} \frac{\sqrt{2x^2+1}}{3x-5}$

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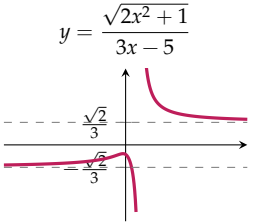
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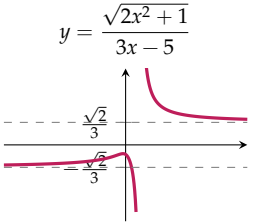
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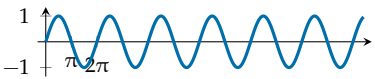
Sanity check: top positive, bottom negative — the limit must be negative.



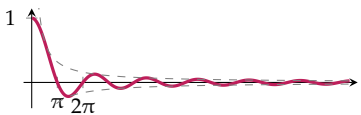
No limit at all, and a limit that is crossed

$\lim_{x\to\infty} \sin x$ does not exist: as x grows, $\sin x$ keeps sweeping through every value in $[-1, 1]$ and never settles. No horizontal asymptote.

$y = \sin x$ — amplitude stays 1



$y = \dfrac{\sin x}{x}$ — amplitude $\to 0$



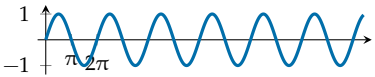
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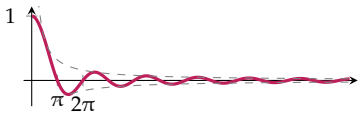
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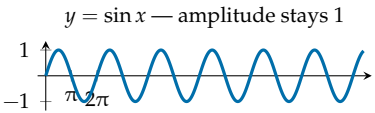
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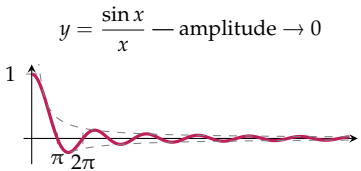
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So $y = 0$ is a horizontal asymptote of $y = \dfrac{\sin x}{x}$, and the graph crosses it at every $x = n\pi$: the warm-up, settled.





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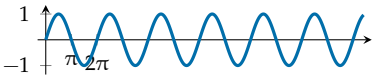
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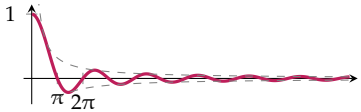
So $y = 0$ is a horizontal asymptote of $y = \dfrac{\sin x}{x}$, and the graph crosses it at every $x = n\pi$: the warm-up, settled.

Compare: $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$. Same function, two ends, two stories.

$y = \sin x$ — amplitude stays 1



$y = \dfrac{\sin x}{x}$ — amplitude $\to 0$



dashed: $y = \pm 1/x$

Infinite limits at infinity

$\lim_{x \to \infty} x^3 = \infty$ and $\lim_{x \to -\infty} x^3 = -\infty$: the values grow without bound — "$= \infty$" is a sentence about behaviour, not a number (Lecture 1).

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Example. $\lim_{x\to\infty} \left(x^2 - x\right)$. Not "$\infty - \infty = 0$"! Factor: $x^2 - x = x\,(x-1)$, both factors $\to \infty$, so the product $\to \infty$.

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Watch the signs: $\lim_{x\to-\infty} \dfrac{x^2+x}{3-x} = +\infty$ (top $\to -\infty$, bottom $\to -1$). The rule of degrees says "$\pm\infty$"; the sign is your job.

A little history · c. 200 BC

The word is Greek: *asymptotos*, "not falling together". Apollonius of Perga used it around 200 BC for the two lines that the hyperbola approaches and never meets — the same hyperbola we met in Lecture 1 among the conic sections. For two thousand years an asymptote was, by definition, a line the curve never touched. The modern definition is about the *limit*, and the curve may cross the line as often as it likes; $\sin x/x$ does so infinitely often. A word can outlive its own definition.

Question 13

Which lines are vertical asymptotes of $y = \dfrac{x-1}{x^2-1}$?

A. $x = 1$ only.

B. $x = -1$ only.

C. Both $x = 1$ and $x = -1$.

D. Neither.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 2 minutes

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120 sec

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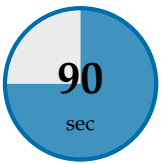
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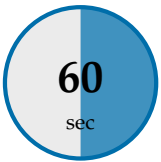
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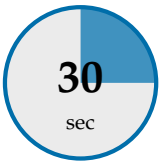
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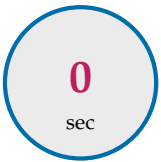
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

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Hands down. Thank you.



Question 13

EXPLAIN

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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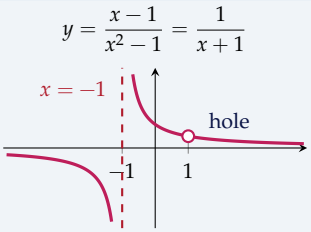
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Answer

B. For $x \neq 1$: $\dfrac{x-1}{(x-1)(x+1)} = \dfrac{1}{x+1}$.

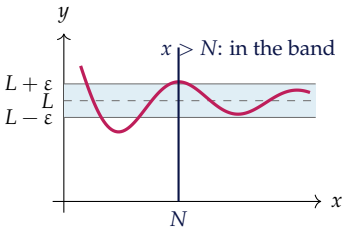
At $x = 1$ the limit is $\frac{1}{2}$: a hole, not an asymptote.
At $x = -1$: $\frac{1}{x+1} \to \pm\infty$: a vertical asymptote.
"Denominator zero" (C) is where to *look*, not the verdict.



The precise definitions

Definition

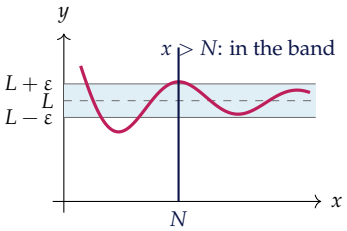
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Compare Lecture 3: "x close to a" ($0 < |x - a| < \delta$) became "x large" ($x > N$); the ε-part is unchanged. For $x \to -\infty$: if $x < N$ then $|f(x) - L| < \varepsilon$.



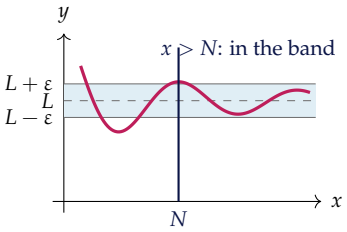
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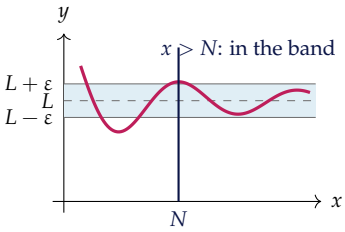
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$\lim_{x\to\infty} f(x) = \infty$ means: for every M there is N such that if $x > N$ then $f(x) > M$.

Same grammar every time: for every challenge there is a response.



A proof from the definition

Claim. $\lim_{x \to \infty} \frac{1}{x} = 0.$

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If $x > N$, then $x > 0$ and $\left|\dfrac{1}{x} - 0\right| = \dfrac{1}{x} < \dfrac{1}{N} = \varepsilon$. $\square$

Two parts, as in Lecture 3: find the response (N) backwards, then write the argument forward. The bigger the challenge (smaller ε), the farther out we must go (larger N) — N depends on ε, as δ did.

"If $f(x) \to \infty$ and $g(x) \to \infty$ as $x \to \infty$, then $f(x) - g(x) \to 0$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

"If $f(x) \to \infty$ and $g(x) \to \infty$ as $x \to \infty$, then $f(x) - g(x) \to 0$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. $x^2 - x \to \infty$; $x - x \to 0$; $(x+1) - x \to 1$; $\sqrt{x^2+x} - x \to \frac{1}{2}$ (Question 12). Both terms go to infinity in every case and the difference does whatever it likes. $\infty - \infty$ is a question, not a number — exactly like $0/0$.

Today in three lines

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This week:

Homework: Stewart §3.4 — read and work out the examples. Next week: the derivative.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"A horizontal asymptote is . . ." (in your own words, no formula).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.